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The dynamics of bla_{TEM} resistance genes in *Salmonella* Typhi

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***Salmonella* Typhi (*S.* Typhi) is an important pathogen causing typhoid fever worldwide. The emergence of antibiotic resistance, including that of bla_{TEM} genes encoding to TEM β-lactamases has been observed. This study aimed to investigate the dynamics of bla_{TEM} genes in *S.* Typhi by analyzing the phylogeny and flanking region patterns and phylogenetic associating them with metadata (year, country) and genomic data (genotypes, antibiotic resistance genes (ARGs), plasmids). Genomic sequences of publicly available *S.* Typhi harboring bla_{TEM} (n = 6079), spanning from 1983 to 2023, were downloaded and analyzed using CSIPhylogeny for phylogeny, Flankophile for identifying genetic contexts around bla_{TEM} genes and GenoTyphi for determining genotypes, ARGs and plasmid replicons. We found that bla_{TEM}-positive isolates occurred most commonly in specific location, especially in Asia and Africa and clustered among a limited number of genotypes. Flankophile identified 740 isolates (12.2%) with distinct flanking region patterns, which were categorized into 13 patterns. Notably, 7 patterns showed a predominantly phylogenetic association with genotypes. Additionally, these 7 patterns exhibited relation to the country, ARGs and plasmid replicons. Further examination of the flanking region patterns provided association with mobile genetic elements (MGEs). Taken together, this study suggests that bla_{TEM} has been acquired by *S.* Typhi isolates a limited number of times and subsequently spread clonally with specific genotypes.**

Keywords *Salmonella* Typhi, Dynamics, Antimicrobial resistance, Antibiotic resistance gene, Flanking region pattern

Salmonella Typhi (*S.* Typhi) remains an important pathogen causing typhoid fever[1]. It causes approximately 11 million cases and 110,000 deaths globally each year[2,3] is particularly a public health concern in low- and middle-income countries[2,4–8], but also in high-income countries remaining relevant due to travel associated infections[1,8–11].

Antimicrobial resistance (AMR) poses a major threat to human and animal health[12]. Studies have reported a widespread emergence in *S.* Typhi isolates[5,6,9,13,14]. Among AMR mechanisms in *S.* Typhi, the production of TEM β-lactamase, expressed from bla_{TEM} gene, plays a crucial role in conferring resistance to β-lactam drugs[9,15,16]. The bla_{TEM} gene family encompasses more than 201 variants[17]. Among these, an important variant reported in *S.* Typhi is bla_{TEM-1B}[18].

Antibiotic resistance genes (ARGs) can be transferred on a large range of mobile genetic elements (MGEs), including plasmids, integrons (Int), insertion sequences (IS) and transposons (Tn)[15]. There is, however, limited knowledge on the dynamics of bla_{TEM} in *S.* Typhi populations. To investigate this, we obtained genomic sequences along with sample metadata from publicly available online sources. Our primary focus was on exploring the dynamics of bla_{TEM} by examining flanking region patterns, the sequences surrounding the targeted resistance gene. We then explored phylogenetic association between these patterns, sample metadata and analyzed genomic data.

Materials and methods

Selection of *S.* Typhi and bla_{TEM} harboring isolates

We gathered *S.* Typhi isolates from three different databases: Pathogenwatch[19] (n = 13,041), Enterobase[20] (n = 11,548), the NCBI Pathogen Detection project[21] (n = 11,688). The metadata from these databases were accessed on 27th November 2023 and combined. Duplicate accession numbers and non-Typhi isolates were excluded and a total of 19,354 unique *S.* Typhi were retrieved for the subsequent selection of bla_{TEM} isolates. Based on the metadata provided these isolates were categorized into: (1) non-bla_{TEM} harboring *S.* Typhi (n = 13,275) and (2) *S.* Typhi harboring bla_{TEM} isolates (n = 6079) that were used for studying of dynamics of bla_{TEM}.

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Genomic sequence downloading and analysis

The genomic data for the 6079 selected *S.* Typhi harboring bla_{TEM} isolates, were downloaded from ENA Portal API. The sequences underwent processing, including pre- and post-trimming quality checks, as well as assembly, using the FoodQCPipeline[22].

Briefly, raw reads were examined for quality using FastQC v0.11.5 (Babraham Bioinformatics, 2016), followed by trimming of sequencing adaptors using bbduk2 (part of BBtools v36.49, https://jgi.doe.gov/data-and-tools/software-tools/bbtools/) with an internal database. Subsequently, the trimmed raw reads were re-evaluated for quality using FastQC. All reads that passed quality were according to: (1) read length was longer or equal to 50 base pairs (bp), (2) phred score per base was higher or equal to 20 and (3) adaptors were removed. The quality-passed reads were then de novo assembled using SPAdes v3.11.0 (Bankevich, et. al., 2012). The resulting assemblies were evaluated for quality using Quast v 4.5 (https://gitlab.univ-lille.fr/mathieu.genete/ngsgenotyp/-/tree/master/TOOLS/quast-4.5).

The confirmation of serovars was conducted using SISTR[23]. The bla_{TEM} variants were confirmed using ABricate v1.0.1 (https://github.com/tseemann/abricate) with three databases, AMRFinderPlus[24], ResFinder[25] and CARD[26]. Genotypes, ARG profiles and plasmid replicon profiles were determined using Genotyphi v2.0[27]. The distribution of the genotypes was visualized using Microreact (https://microreact.org/project/6aHVsxbEyr1WKsFjqM9wgC-styphiblatemv1)[28].

Phylogenetic tree construction

The assembled genomes of the 6079 *S.* Typhi harboring bla_{TEM} were aligned to the reference genome of *S.* Typhi CT18, an XDR strain from Vietnam in the year 1993 (accession no: AL513382) using CSIPhylogeny pipeline[29]. The resulting core SNP alignment was used for inferring a maximum-likelihood tree with Fastree v2.1.3[30]. The phylogenetic tree (SNP Tree), sample metadata and analyzed genomic results were then visualized by Microreact[28].

Generating bla_{TEM} gene synteny

The bla_{TEM} gene synteny was analyzed using Flankophile v0.2.7, a tool for analyzing and visualization of genetic context surrounding the targeted resistance gene (gene synteny). The contigs carrying bla_{TEM} of the 6079 selected *S.* Typhi isolates were used as an input. Concerning Flankophile configurations: (1) ResFinder (08/02/2022) database was utilized with a minimum coverage and identity to the reference sequence of > 98%, (2) flanking length (both upstream and downstream) was defined as 2000 base pairs (bp), (3) cluster identity and length differences were set at 0.98, (4) k-mer size indexing was set at 16 and (5) distance matrix calculation was conducted using method number one which involves k-mer hamming distance[31]. The Flankophile outputs were analyzed and reported as the flanking region patterns.

Results

Temporal and geographical distribution of *Salmonella* Typhi isolates

All bla_{TEM} harboring genomic sequences were downloaded, re-confirmed serovar Typhi using SISTR and classified as *S.* Typhi harboring bla_{TEM} isolates (n = 6079; 31.4%). While isolates that did not harbor bla_{TEM} were classified as non-bla_{TEM} harboring *S.* Typhi (n = 13,275; 68.6%) (Fig. 1 & Supplementary Table 1). Most bla_{TEM} variant were $bla_{\mathrm{TEM\text{-}1B}}$ (n = 6037), but we also found $bla_{\mathrm{TEM\text{-}135}}$ (n = 8), $bla_{\mathrm{TEM\text{-}116}}$ (n = 1), $bla_{\mathrm{TEM\text{-}215}}$ (n = 1) and $bla_{\mathrm{TEM\text{-}others}}$ (n = 32). Among all bla_{TEM}-positive isolates, there were 851 isolates showing co-existence with ESBL genes. One of the most common ESBL genes was $bla_{\mathrm{CTX\text{-}M\text{-}15}}$ β-lactamase gene (n = 834/851) (Supplementary Table 2).

Our dataset revealed information of *S.* Typhi isolates, spanning from 1916 to 2023 with the first bla_{TEM}-positive isolates observed from Chile in 1983 (Supplementary Table 3). Although the number of bla_{TEM} harboring isolates has been increasing based on sequencing input, the proportion of bla_{TEM}-positive isolates remained relatively stable over time. The bla_{TEM}-positive isolates were predominantly from Asia (Southern, South-eastern) and Africa (Eastern, Western and Southern) and had a more geographically restricted distribution than bla_{TEM}-negative isolates (Fig. 1).

Association of bla_{TEM} to genotypes

The distribution of bla_{TEM}-positive *S.* Typhi isolates was also compared to genotypes. The bla_{TEM}-negative isolates exhibited greater genotypic diversity of genotype than the bla_{TEM}-positive isolates and the later isolates were mainly found in genotype 4.3.1.1, 4.3.1.1.EA1, 4.3.1.1.P1, 3.1.1, 4.3.1.3.Bdq and and 2.5.1, mainly from Asia and Africa (Fig. 2 and Supplementary Fig. 1 & Supplementary Table 4).

Associations between phylogeny, genotypes, plasmid and resistance profiles of bla_{TEM} positive isolates

The 6079 *S.* Typhi harboring bla_{TEM} isolates were used for constructing maximum-likelihood phylogenetic tree. SNP-analyses showed large genomic diversity among the bla_{TEM}-positive isolates, with the phylogenetic tree in general clustering with the defined genotypes (Supplementary Fig. 2). However, some discrepancies were also observed. Thus, especially genotype 4.3.1.1 clustered in groups into different parts of the phylogeny, genotypes 2.3.1, 2.3.4, 2.5.1, 3, 3.1.1 and 3.2.1 closely together, as did genotypes 4.3.1, 4.3.1.2, 4.3.1.2.EA2, 4.3.2.2.EA3, 4.3.1.3 and 4.3.1.3.Bdq. There was also some clustering of isolates from the same countries, but in several cases, clones were observed across several countries and even continents. The antibiotic resistance gene (ARG) profile also showed association to phylogeny. Almost all bla_{TEM}-positive isolates also harbored *catA1*, *dfrA7*, *sul1* and *sul2*. However, exceptions were observed (Supplementary Fig. 2).

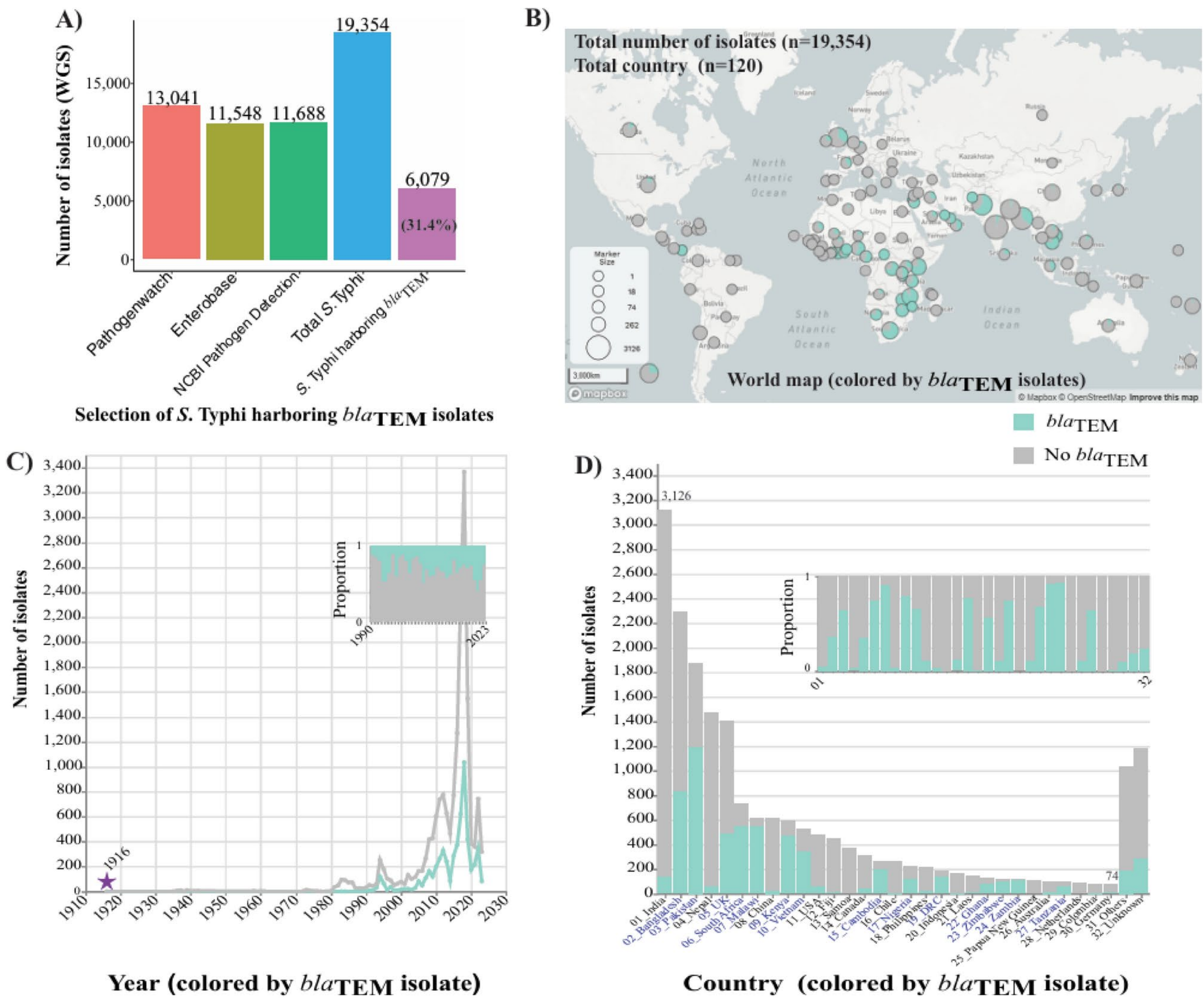

Fig. 1. Acquisition and distribution of *Salmonella* Typhi isolates (harboring vs non-harboring *bla*$_{TEM}$). Acquisition of *S.* Typhi harboring *bla*$_{TEM}$ isolates from publicly available online database (**A**). The distribution of *S.* Typhi isolates (harboring vs non-harboring *bla*$_{TEM}$) shows as the world map (**B**), year of isolation **(C)** and country of origin (**D**). Legend, green is *S.* Typhi harboring *bla*$_{TEM}$ isolates (n = 6079) and grey is non-harboring *bla*$_{TEM}$ isolates (n = 13,275).

A limited number of isolates harbored *bla*$_{CTX\text{-}M\text{-}15}$, but even though this was identified in two different resistance profiles and across the different phylogenetic clusters, no clear association to plasmids was observed. One exception was a very clear association between resistance profile 10 (negative for *catA1*, *tet*(B) and *sul2*) and plasmid profile 6 (IncFIB(K) among isolates from Bangladesh (Supplementary Fig. 2). All ARG profiles were classified in to 16 profiles (Supplementary Fig. 3 and Supplementary Table 5). All plasmid profiles were classified in to 13 plasmid profiles (Supplementary Table 6).

Analysis of *bla*$_{TEM}$ flanking region patterns

It was only possible to obtain sufficient flanking regions (2000 bp on each side) for 740 isolates (12.2%) (Supplementary Fig. 4 and Supplementary Table 7). Those isolates were not equally distributed across the phylogenetic tree, but were mainly from genotype 3.1.1 in Nigeria and Ghana, 4.3.1.3.Bdq in Bangladesh, 4.3.1.1.P1 in Pakistan and 4.3.1.1.EA1 in Zambia and Tanzania (Fig. 3 and Supplementary Figs. 5 and 6). Flanks of 5000 bp were available from 577 isolates (Supplementary Fig. 7 and Supplementary Table 8).

The flankophile outputs were then categorized into 13 different patterns (01–13). Among them, there were 7 dominant patterns, namely pattern 01 (n = 240), pattern 02 (n = 167), pattern 03 (n = 48), pattern 04 (n = 22), pattern 05 (n = 128), pattern 06 (n = 19) and pattern 07 (n = 80) (Supplementary Fig. 4 and Supplementary Table 7).

The seven dominant flanking patterns (01–07) were closely linked to genotype, ARG profile and plasmid profile, but less to country of origin and isolation year (Fig. 3 and Supplementary Figs. 5 and 6).

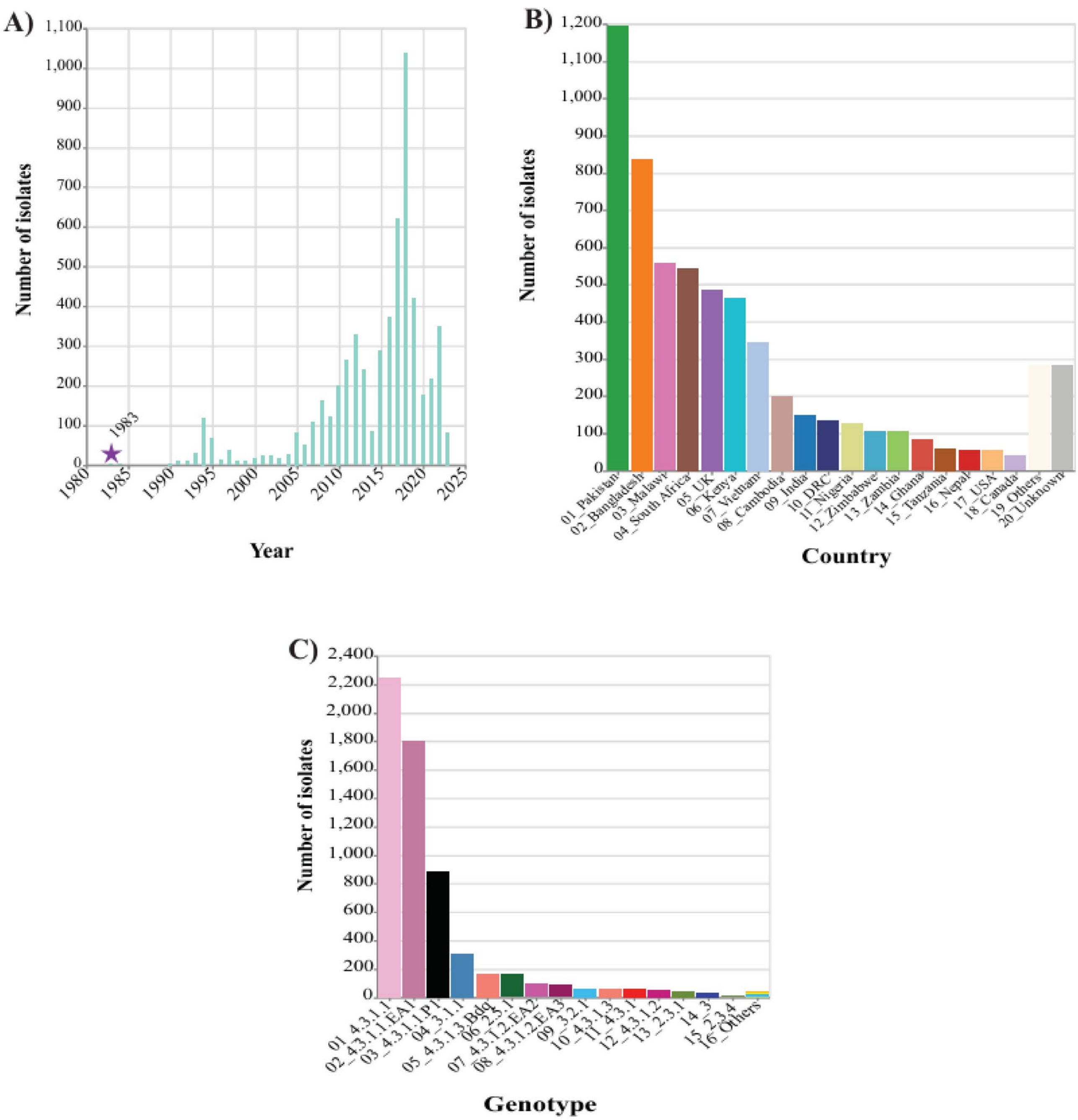

Fig. 2. Distribution of *Salmonella* Typhi harboring bla_{TEM} isolates. The distribution of the selected isolates (n = 6079) is divided by year (**A**), country (**B**) and genotype (**C**).

Flanking region pattern (Pattern) 01 clustered with genotype 3.1.1-mainly isolated from Nigeria and Ghana, plasmid profile 5 and ARG profile 5. However, Pattern 03 clustered phylogenetically here and with a different plasmid profile namely IncY (profile 1) and ARG profile 14. Pattern 03 was only linked with genotype 3.1.1 and was mainly isolated from Nigeria. However, in 3 isolates the pattern was linked to USA (n = 2) and UK (n = 1) (Fig. 3 and Supplementary Figs. 5 and 6).

Pattern 02 was associated with 4.3.1.3.Bdq, mainly originated from Bangladesh and IncFIB(K) (plasmid profile 6) and ARG profile 10. Pattern 04 was observed among 22 isolates and distributed quite broadly across the phylogeny, but associated to plasmid profile 7 (IncN) (Fig. 3 and Supplementary Figs. 5 and 6).

Pattern 05 clustered with genotype 4.3.1.1.P1, was mainly isolated from Pakistan and associated with ARG profile 4 and plasmid pattern 1 (IncY). Pattern 06 was also related to genotype 4.3.1.1.P1, ARG profile 4 and plasmid profile 1 and was observed from various countries, Pakistan (n = 5), UK (n = 5), USA (n = 3). Pattern 07 clustered with 4.3.1.1.EA1, was isolated from Africa (Zambia and Tanzania), mainly associated to ARG profile 15 and plasmid profile 6 (IncFIB(K) (Fig. 3 and Supplementary Figs. 5 and 6).

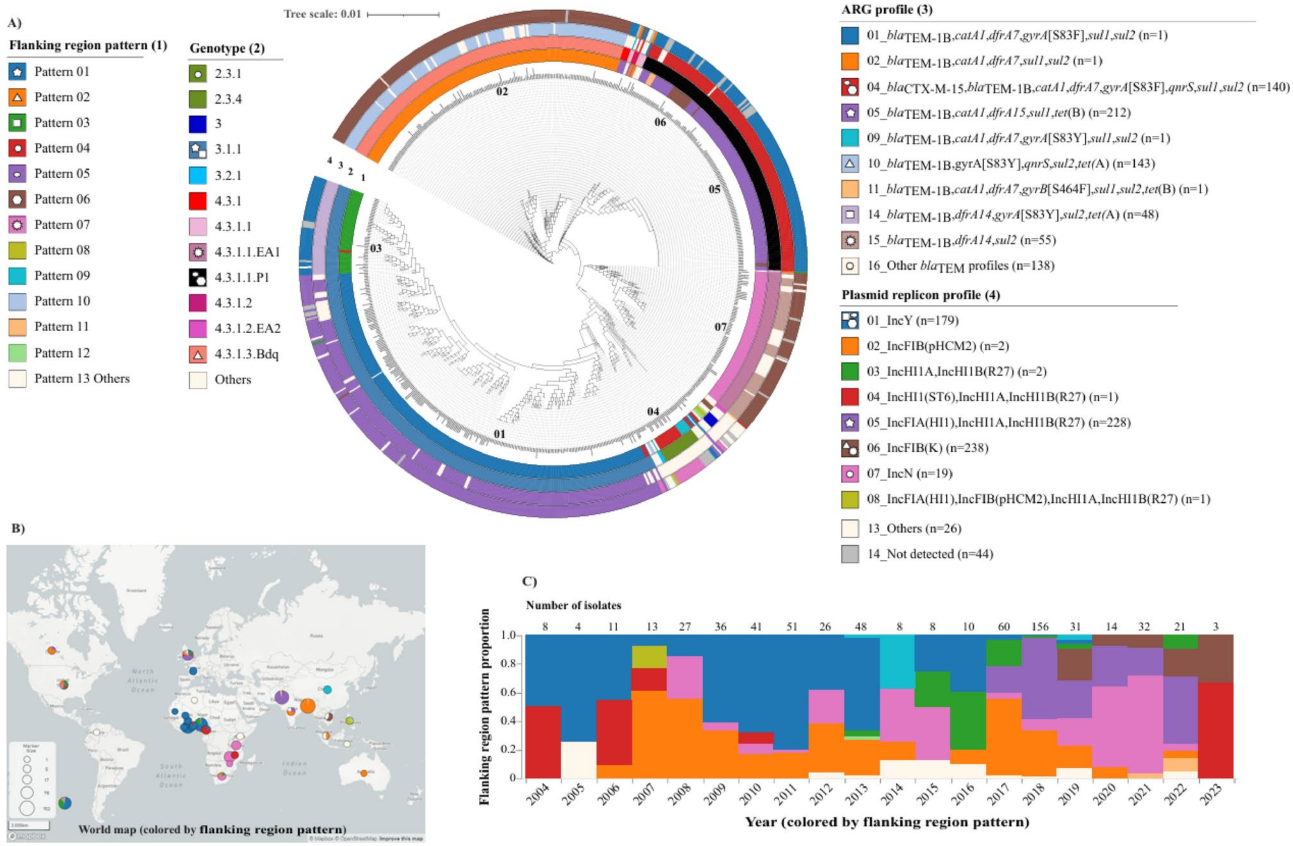

Fig. 3. Association of flanking region patterns with analyzed genomic data. The flanking region patterns (1) are linked to genotype (2), antibiotic resistance gene (ARG) profile (3) and plasmid replicon profile (4) (**A**). The *S.* Typhi harboring *bla*$_{TEM}$ isolates with flankophile outputs (n = 740) are used for constructing a SNP Tree using CT18 as reference strain. The tree has 36,350 SNP and 76.5% coverage. The distribution of flanking region pattern of *S.* Typhi harboring *bla*$_{TEM}$ isolates shows as the world map (**B**). The dynamics of flanking region patterns is shown between 2004 and 2023 (**C**). The symbols of seven common flanking region patterns, pattern 01 (five point star), pattern 02 (triangle), pattern 03 (square), pattern 04 (circle), pattern 05 (oval), pattern 06 (hexagon) and pattern 07 (multipoint star), depict linkage of the flanking region pattern with other metadata.

The major clustering of SNP-phylogeny and flanking region pattern is also confirmed and illustrated using tanglegram (Fig. 4).

We further analyzed the genetic contexts surrounding *bla*$_{TEM}$, specifically focusing on insertion sequences, within a 5000 bp flanking region using the Flankophile tool. Pattern 01 was included upstream Tn2 (n = 240). Pattern 02 was contained upstream Tn2 and downstream ISVsa3 and IS5075 (n = 160). Pattern 03 was contained upstream ISAs25 and ISEc63 (n = 47). Pattern 04 was included upstream Tn2 (n = 20). Pattern 05 was comprised upstream ISEcp1 and downstream IS5075 (n = 11). Pattern 07 was upstream ISEc63 and downstream IS5075 (n = 75). Overall, Tn2 emerged as the most common upstream element, while IS5075 was identified as the dominant downstream element (Supplementary Fig. 7 and Supplementary Table 8).

All flanking region patterns were illustrated individually for clustering with metadata profile (genotype, ARG profile, plasmid profile and country) (Supplementary Figs. 8–20).

Discussion

S. Typhi harboring *bla*$_{TEM}$

Our study focused on *S.* Typhi harboring *bla*$_{TEM}$ isolates obtained from publicly available data. Major geographical differences in *S.* Typhi harboring *bla*$_{TEM}$ were found and a high prevalence of the *bla*$_{TEM}$ isolates was dominantly observed in Asia and Africa. Despite the presence of surveillance programs, the number of reported cases remains underestimated, particularly in developing countries. Enhanced surveillance and accurate reporting are essential for determining the true distribution of resistance and for collecting data to inform public health decisions and policies. The *bla*$_{TEM-1B}$ was the most common variant found in the *S.* Typhi harboring *bla*$_{TEM}$ isolates (99.3%), and only a few other variants (*bla*$_{TEM-135}$, *bla*$_{TEM-215}$) were observed. The co-existence of *bla*$_{TEM-1B}$ and *bla*$_{CTX-M-15}$ (n = 834), encoding resistance to cephalosporins, was predominantly observed (13.7%). This result is consistent with several previous reports on high prevalence of *bla*$_{TEM}$ genes in these regions[2,32]. The isolates carrying the *bla*$_{CTX-M-15}$ gene exhibited XDR properties[5].

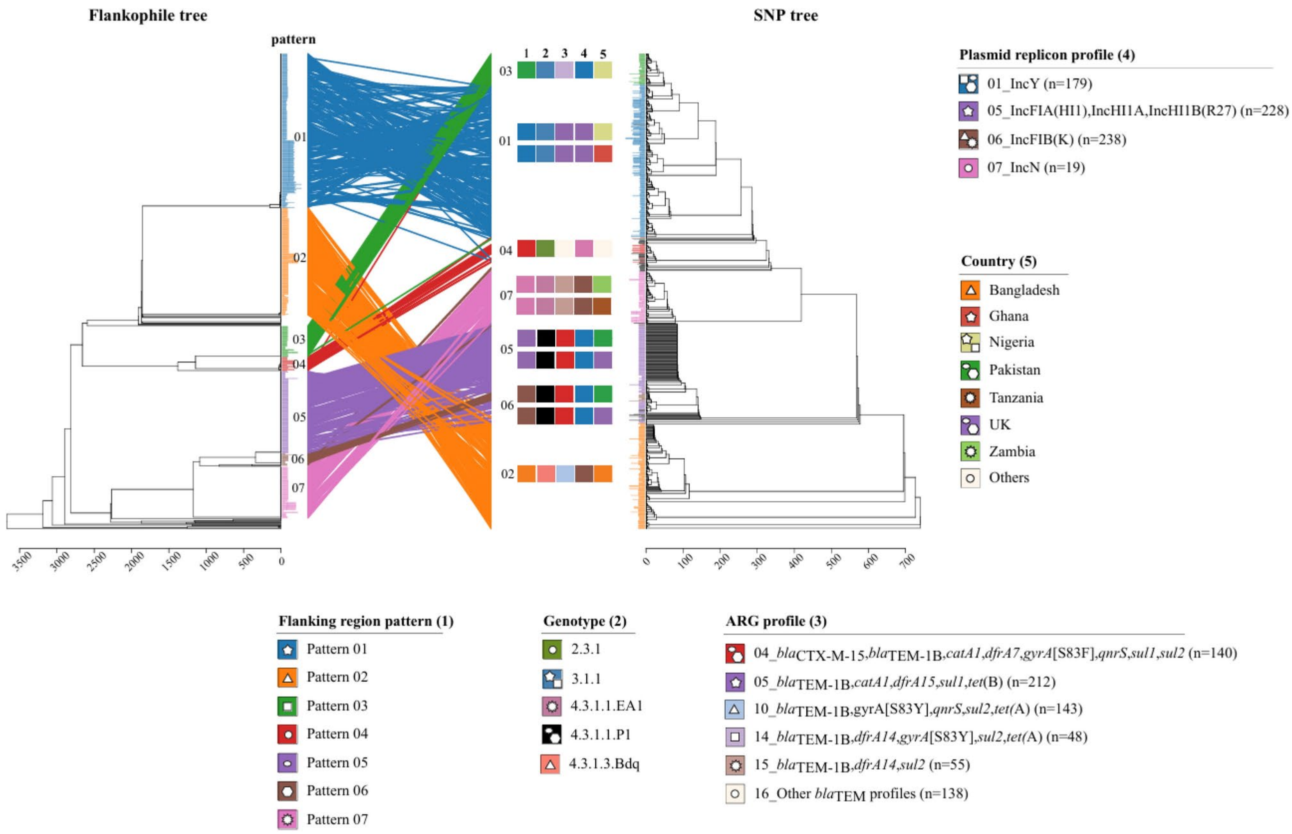

Fig. 4. Clustering of flanking region patterns of *S*. Typhi harboring bla_{TEM} isolates. The clustering is linked between a flankophile tree constructed using Flankophile and a SNP tree constructed using CSIPhylogeny. The clustering represents association between flanking region pattern (1) and genotype (2), ARG profile (3), plasmid replicon profile (4) and country (5). The symbols of seven common flanking region patterns, pattern 01 (five point star), pattern 02 (triangle), pattern 03 (square), pattern 04 (circle), pattern 05 (oval), pattern 06 (hexagon) and pattern 07 (multipoint star), depict linkage of the flanking region pattern with other metadata.

Genotyping is a valuable approach for examining local *S*. Typhi populations and detecting recent introductions of specific genotypes in new or re-endemic areas[33,34]. *S*. Typhi harboring bla_{TEM} isolates were occurred in a limit number of genotypes, especially genotypes 4.3.1.1, 4.3.1.1.P1, 4.3.1.3.Bdq, 4.3.1.1.EA1, 3.1.1 and 2.5.1. While non-bla_{TEM} isolates varied much more in genotypes and geographical isolation location.

In general, a strong correlation between genotypes, ARG profile and plasmid profiles were observed, suggesting that bla_{TEM} have not emerged multiple times among several clones of *S*. Typhi, but more likely that acquisition has only occurred a few times followed by subsequent spread of resistant variants. *S*. Typhi harboring bla_{TEM} mainly belonged to genotypes 4.3.1 and its sublineages. This result is consistent with a report by Silva et al., which estimated that the first transfer of 4.3.1 (H58) originated from India and subsequently spread internationally and intercontinentally[35]. Genotype 4.3.1 (H58) has recently disseminated across different continents[34]. It was commonly reported in Asia (Bangladesh, Pakistan, Nepal and India)[6,7], Africa (Kenya, East, West)[10,36,37] and individuals returning from endemic countries, including the UK and Australia[1,9].

Dynamics of flanking region patterns

We also performed an analysis of flanking regions of the bla_{TEM} genes. This could only be done for a subset of the data due to the unequal quality of the sequencing data. However, we found some common MGEs, Tn2, ISEc63, ISEcp1 and IS5075. This suggests that bla_{TEM}, surrounded by mobile genetic elements (MGEs), particularly Tn2, has the potential for transmission and altering the genes surrounding the bla_{TEM} gene. The flanking region analyses in general confirmed that bla_{TEM} has been acquired a limited number of times followed by subsequent spread of the new resistant clones. However, a few cases of possible horizontal transmission were also found.

This study is the first largest analysis of flanking regions of bla_{TEM}-positive isolates and flanking region analyses thus supports the initial findings, but we also observed that a better quality of data is needed to perform such analyses.

Limitations and suggestions

The sample metadata lacking information on serovars or ARGs was excluded, potentially leading to an underestimation of the total number of eligible isolates. In the flankophile output, only 740 isolates (12.2%) successfully passed the default setting with a flanking length of 2000 bp. Exclusions were based on contig length and the percentage of identity and coverage (% < 98). Therefore, enhancing sequencing quality by obtaining longer contigs or utilizing sequences from long-read platforms could significantly improve the genomic information available for studying the evolution and dynamics of targeted resistance genes using the Flankophile tool.

Conclusion

Our analysis of publicly available *S.* Typhi genomes showed bla_{TEM} harboring isolates were most commonly found in Asia and Africa and distributed in a limit number of genotypes, especially 4.3.1 and its sublineages. The bla_{TEM} flanking region patterns were dominantly associated to genotypes and associated with ARG profiles, plasmid replicons and countries. Our study suggests that bla_{TEM} has been independently acquired by a limited number of *S.* Typhi genotypes and likely with subsequent spread of the resistant variants regionally. However, while clonal transmission of resistant variants seems to be the most important reason for bla_{TEM} emergence, a few cases of horizontal transmission between genotypes was also observed.

Data availability

All metadata used in this study can be searched at: https://microreact.org/project/6aHVsxbEyr1WKsFjqM-9wgC-styphiblatemv1.

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References

1. Ingle, D. J. *et al.* Genomic epidemiology and antimicrobial resistance mechanisms of imported typhoid in Australia. *Antimicrob. Agents Chemother.* **65**, e0120021. https://doi.org/10.1128/aac.01200-21 (2021).
2. Stanaway, J. D. *et al.* The global burden of typhoid and paratyphoid fevers: A systematic analysis for the global burden of disease study 2017. *Lancet Infect. Dis.* **19**, 369–381. https://doi.org/10.1016/s1473-3099(18)30685-6 (2019).
3. Antillón, M. *et al.* The burden of typhoid fever in low- and middle-income countries: A meta-regression approach. *PLoS Negl. Trop. Dis.* **11**, e0005376. https://doi.org/10.1371/journal.pntd.0005376 (2017).
4. Lima, N. C. B. *et al.* Analysis of isolates from Bangladesh highlights multiple ways to carry resistance genes in *Salmonella* Typhi. *BMC Genomics* **20**, 530. https://doi.org/10.1186/s12864-019-5916-6 (2019).
5. Jacob, J. J. *et al. Salmonella* Typhi acquires diverse plasmids from other Enterobacteriaceae to develop cephalosporin resistance. *Genomics* **113**, 2171–2176. https://doi.org/10.1016/j.ygeno.2021.05.003 (2021).
6. Tanmoy, A. M. *et al. Salmonella* enterica serovar Typhi in Bangladesh: Exploration of genomic diversity and antimicrobial resistance. *mBio* **9**. https://doi.org/10.1128/mBio.02112-18 (2018).
7. Pragasam, A. K. *et al.* Phylogenetic analysis indicates a longer term presence of the globally distributed H58 haplotype of *Salmonella* Typhi in Southern India. *Clin. Infect. Dis.* **71**, 1856–1863. https://doi.org/10.1093/cid/ciz1112 (2020).
8. Als, D. *et al.* Global trends in typhoidal salmonellosis: A systematic review. *Am J Trop Med Hyg* **99**, 10–19. https://doi.org/10.4269/ajtmh.18-0034 (2018).
9. Chattaway, M. A. *et al.* Phylogenomics and antimicrobial resistance of *Salmonella* Typhi and Paratyphi A, B and C in England, 2016–2019. *Microb. Genom.* **7**. https://doi.org/10.1099/mgen.0.000633 (2021).
10. Ingle, D. J. *et al.* Informal genomic surveillance of regional distribution of *Salmonella* Typhi genotypes and antimicrobial resistance via returning travellers. *PLoS Negl. Trop. Dis.* **13**, e0007620. https://doi.org/10.1371/journal.pntd.0007620 (2019).
11. Bokhary, H., Pangesti, K. N. A., Rashid, H., Ghany, M. A. E. & Hill-Cawthorne, G. A. Travel-related antimicrobial resistance: A systematic review. *Trop. Med. Infect. Dis.* **6**. https://doi.org/10.3390/tropicalmed6010011 (2021).
12. Yusof, N. Y. *et al.* Prevalence of antimicrobial resistance genes in *Salmonella* Typhi: A systematic review and meta-analysis. *Trop. Med. Infect. Dis.* **7**. https://doi.org/10.3390/tropicalmed7100271 (2022).
13. Wong, V. K. *et al.* Phylogeographical analysis of the dominant multidrug-resistant H58 clade of *Salmonella* Typhi identifies inter- and intracontinental transmission events. *Nat. Genet.* **47**, 632–639. https://doi.org/10.1038/ng.3281 (2015).
14. Yan, M. *et al.* The emergence and outbreak of multidrug-resistant typhoid fever in China. *Emerg. Microbes Infect.* **5**, e62. https://doi.org/10.1038/emi.2016.62 (2016).
15. Singh, N. S., Singhal, N. & Virdi , J. S. Genetic environment of bla_{TEM-1}, $bla_{CTX-M-15}$, bla_{CMY-42} and Characterization of Integrons of *Escherichia coli* Isolated From an Indian Urban Aquatic Environment. *Front. Microbiol.* **9**. https://doi.org/10.3389/fmicb.2018.00382 (2018).
16. Nuanmuang, N., Leekitcharoenphon, P., Njage, P. M. K., Gmeiner, A. & Aarestrup, F. M. An overview of antimicrobial resistance profiles of publicly available *Salmonella* genomes with sufficient quality and metadata. *Foodborne Pathog. Dis.* **20**, 405–413. https://doi.org/10.1089/fpd.2022.0080 (2023).
17. NCBI. Reference gene catalog: gene_family: bla_{TEM}. Available from: https://www.ncbi.nlm.nih.gov/pathogens/refgene/#gene_family:blaTEM [Last accessed:July/10/2023]. (2023).
18. Wang, Y. *et al.* Extensively drug-resistant (XDR) *Salmonella* Typhi outbreak by waterborne infection - Beijing municipality, China, January-February 2022. *China CDC Wkly* **4**, 254–258. https://doi.org/10.46234/ccdcw2022.062 (2022).
19. Argimón, S. *et al.* A global resource for genomic predictions of antimicrobial resistance and surveillance of *Salmonella* Typhi at pathogenwatch. *Nat. Commun.* **12**, 2879. https://doi.org/10.1038/s41467-021-23091-2 (2021).
20. Alikhan, N. F., Zhou, Z., Sergeant, M. J. & Achtman, M. A genomic overview of the population structure of *Salmonella*. *PLoS Genet.* **14**, e1007261. https://doi.org/10.1371/journal.pgen.1007261 (2018).
21. Sayers, E. W. *et al.* Database resources of the national center for biotechnology information. *Nucleic Acids Res* **50**, D20-d26. https://doi.org/10.1093/nar/gkab1112 (2022).
22. CGE. FoodQCPipeline; 2016. https://bitbucket.org/genomicepidemiology/foodqcpipeline/src/master/. [Last accessed: July/15/2022]. (2016).
23. Yoshida, C. E. *et al.* The *Salmonella* in silico typing resource (SISTR): An open web-accessible tool for rapidly typing and subtyping draft *Salmonella* genome assemblies. *PLoS One* **11**, e0147101. https://doi.org/10.1371/journal.pone.0147101 (2016).
24. Feldgarden, M. *et al.* Validating the AMRFinder tool and resistance gene database by using antimicrobial resistance genotype-phenotype correlations in a collection of isolates. *Antimicrob. Agents Chemother.* https://doi.org/10.1128/aac.00483-19 (2019).
25. Bortolaia, V. *et al.* ResFinder 4.0 for predictions of phenotypes from genotypes. *J. Antimicrob. Chemother.* **75**, 3491–3500. https://doi.org/10.1093/jac/dkaa345 (2020).
26. Jia, B. *et al.* CARD 2017: expansion and model-centric curation of the comprehensive antibiotic resistance database. *Nucleic Acids Res.* **45**, D566–D573. https://doi.org/10.1093/nar/gkw1004 (2017).

27. Dyson, Z. A. & Holt, K. E. Five Years of GenoTyphi: Updates to the global *Salmonella* Typhi genotyping framework. *J. Infect. Dis.* **224**, S775–S780. https://doi.org/10.1093/infdis/jiab414 (2021).
28. Argimón, S. *et al.* Microreact: visualizing and sharing data for genomic epidemiology and phylogeography. *Microb. Genom.* **2**, e000093. https://doi.org/10.1099/mgen.0.000093 (2016).
29. Kaas, R. S., Leekitcharoenphon, P., Aarestrup, F. M. & Lund, O. Solving the problem of comparing whole bacterial genomes across different sequencing platforms. *PLoS One* **9**, e104984. https://doi.org/10.1371/journal.pone.0104984 (2014).
30. Price, M. N., Dehal, P. S. & Arkin, A. P. FastTree 2–approximately maximum-likelihood trees for large alignments. *PLoS One* **5**, e9490. https://doi.org/10.1371/journal.pone.0009490 (2010).
31. Thorn, A. V., Aarestrup, F. M. & Munk, P. Flankophile: A bioinformatic pipeline for prokaryotic genomic synteny analysis. *Microbiol. Spectr.* **12**, e0241323. https://doi.org/10.1128/spectrum.02413-23 (2024).
32. Carey, M. E. *et al.* Global diversity and antimicrobial resistance of typhoid fever pathogens: Insights from a meta-analysis of 13,000 *Salmonella* Typhi genomes. *medRxiv* https://doi.org/10.1101/2022.12.28.22283969 (2023).
33. Wong, V. K. *et al.* An extended genotyping framework for *Salmonella enterica* serovar Typhi, the cause of human typhoid. *Nat. Commun.* **7**, 12827. https://doi.org/10.1038/ncomms12827 (2016).
34. Dyson, Z. A. & Holt, K. E. Five years of GenoTyphi: Updates to the global *Salmonella* Typhi genotyping framework. *J. Infect. Dis.* **224**, S775–S780. https://doi.org/10.1093/infdis/jiab414 (2021).
35. da Silva, K. E. *et al.* The international and intercontinental spread and expansion of antimicrobial-resistant *Salmonella* Typhi: A genomic epidemiology study. *Lancet Microbe* **3**, e567–e577. https://doi.org/10.1016/S2666-5247(22)00093-3 (2022).
36. Kariuki, S. *et al.* Multiple introductions of multidrug-resistant typhoid associated with acute infection and asymptomatic carriage, Kenya. *Elife* https://doi.org/10.7554/eLife.67852 (2021).
37. Ochieng, C. *et al.* Molecular characterization of circulating *Salmonella* Typhi strains in an urban informal settlement in Kenya. *PLoS Negl. Trop. Dis.* **16**, e0010704. https://doi.org/10.1371/journal.pntd.0010704 (2022).

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Author contributions
F.M.A. and P.L. conceptualized the study. N.N. downloaded and analyzed data, visualized the results and wrote the first draft. N.N., F.M.A., P.L., P.M.K.N., A.V.T. participated in the result analysis and discussion. N.N., F.M.A., P.L., P.M.K.N., A.V.T. edited the final version of the article. All authors contributed to reviewing and approving the final article.

Declarations

Competing interests
The authors declare no competing interests.

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